Lecture-19

Revanth Reddy Pannala
EBIT, Unibo

→ we studied in the last class the combination of current Mirrors and differential pairs as a mean to achieve "single ended Output"

and also analyzed the ICMR for the current Mirror loaded Differential Amplifier.

This Analysis was regarding Biasing and determining the operating point of the Circuit

→ But we are also interested in finding the Gain of the "single Ended Differential Amplifiers"

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single-ended differential amplifiers

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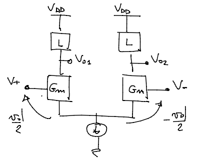
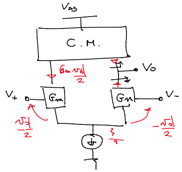
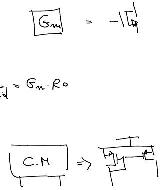
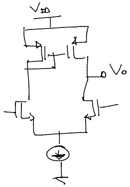
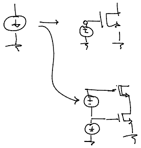


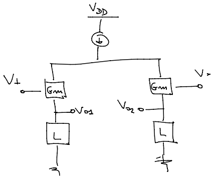


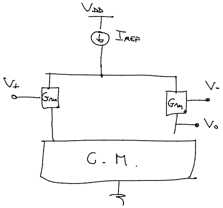


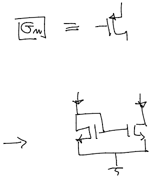
$$\frac{v_o}{v_{id}} = G_m \cdot R_o$$

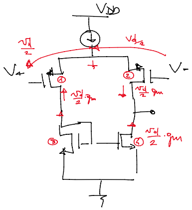




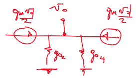


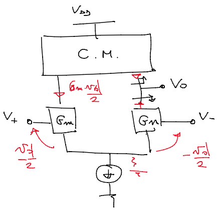




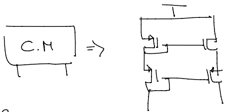


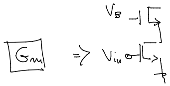
$$\frac{v_o}{v_{id}} = \frac{g_m}{g_{o2} + g_{o4}}$$

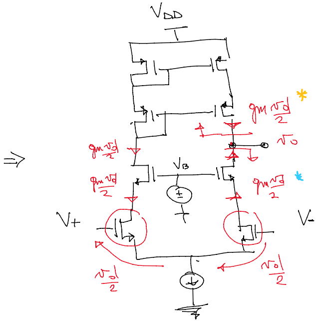


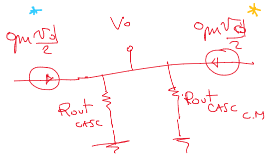


$$\frac{v_o}{v_{id}} = G_m \cdot R_o$$



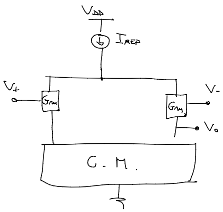


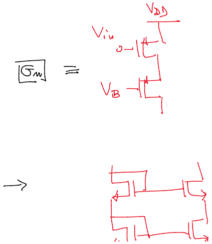


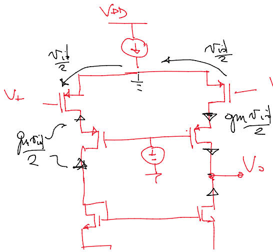


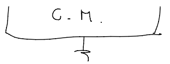
$$v_o = g_m v_d \cdot R_{out}$$

$$\frac{v_o}{v_d} = g_m \cdot R_{out\,(CASC \,||\, CASC\,CM)}$$









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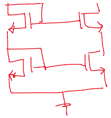
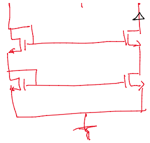
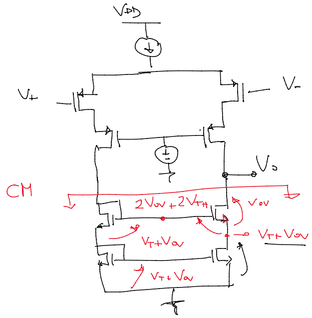


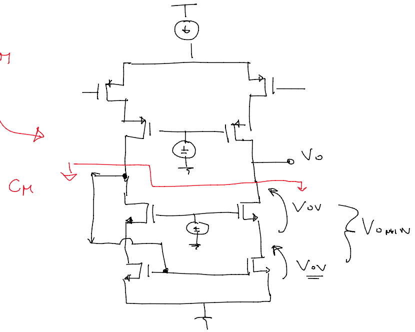


Cascode CM

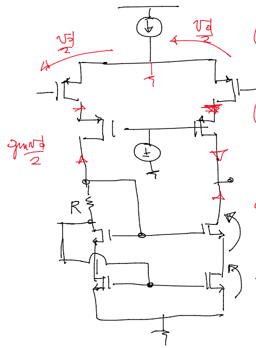
→ High Swing Cascode CM



$V_{OMIN} = 2V_o + V_T$



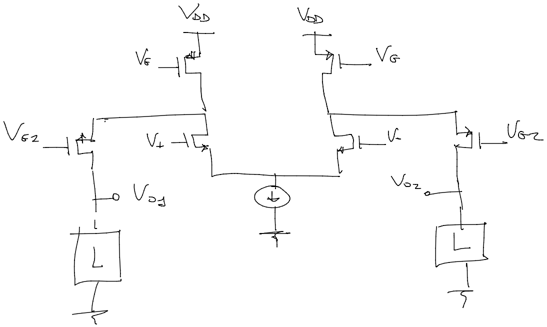
Self biased Cascode



$V_{OMIN} = 2V_{ov}$

Folded Cascode

Diff. Output

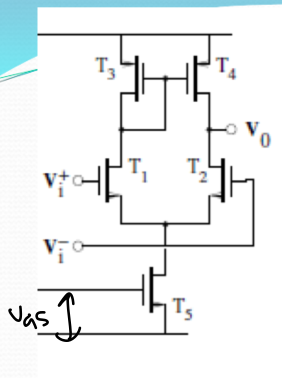


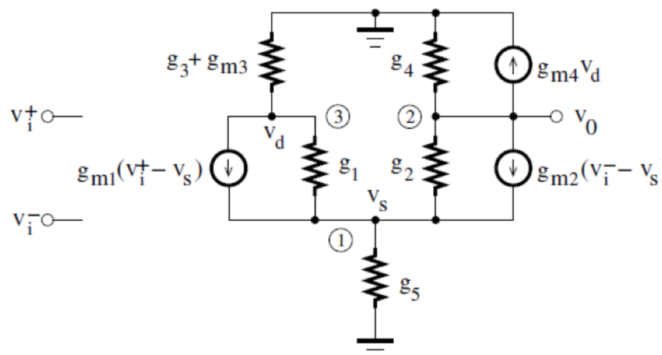
Folded Cascode

Single-ended Output

?

* Small Signal Analysis of the Single Ended differential Amplifier *





$$\text{(1)}\quad g_{m1}(v_i^+ - v_s) + g_1(v_d - v_s) + g_{m2}(v_i^- - v_s) + g_2(v_0 - v_s) - g_5 v_s = 0,$$

$$\text{(2)}\quad g_{m2}(v_i^- - v_s) + g_2(v_0 - v_s) + g_{m4}v_d + g_4 v_0 = 0,$$

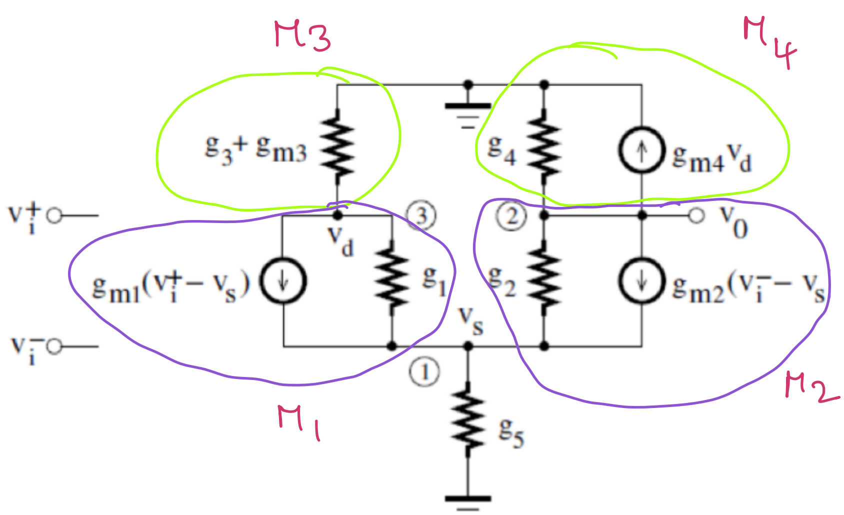
$$\text{(3)}\quad (g_3 + g_{m3})v_d + g_{m1}(v_i^+ - v_s) + g_1(v_d - v_s) = 0$$

where $g_1 = g_{ds_1}$, $g_3 = g_{ds_3}$, and $g_5 = g_{ds_5}$

$g_{m1} = g_{m2}$, $g_1 = g_2$, $g_{m3} = g_{m4}$, and $g_3 = g_4$

$$\boxed{v_{id} = v_i^+ - v_i^- \qquad v_{ic} = \frac{v_i^+ + v_i^-}{2}.}$$

→



$M_1, M_2 \rightarrow$ Transistor of Differential pair in SSM

$M_3 \Rightarrow$ Diode connected Load. SSM

$M_4 \rightarrow$ PMOS SSM

→ In all case typical "Back Gate Transconductance" is neglected.

$$2g_{m1}v_{ic} + (g_1 + g_3)v_0 + (g_1 + g_3 + 2g_{m3})v_d - 2(g_1 + g_{m1})v_s = 0$$

$$2g_{m1}v_{ic} + g_1 v_0 + g_1 v_d - [g_5 + 2(g_1 + g_{m1})]v_s = 0,$$

$$g_{m1}v_{id} - (g_1 + g_3)v_0 + (g_1 + g_3)v_d = 0,$$

$$v_{id} = v_i^+ - v_i^-$$

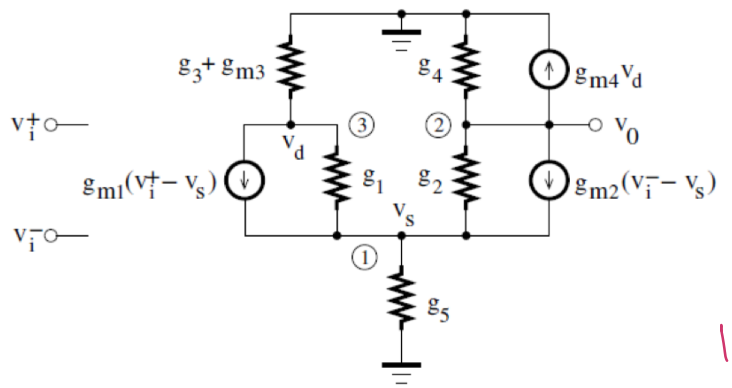
$$v_{ic} = \frac{v_i^+ + v_i^-}{2}.$$

$$v_0 = A_d v_{id} + A_c v_{ic}$$

$$A_d = \frac{g_{m1}\{g_1 g_5 + (g_3 + 2g_{m3})[g_5 + 2(g_1 + g_{m1})]\}}{2(g_1 + g_3)\{g_1 g_5 + (g_3 + g_{m3})[g_5 + 2(g_1 + g_{m1})]\}} \simeq \frac{g_{m1}}{g_1 + g_3}$$

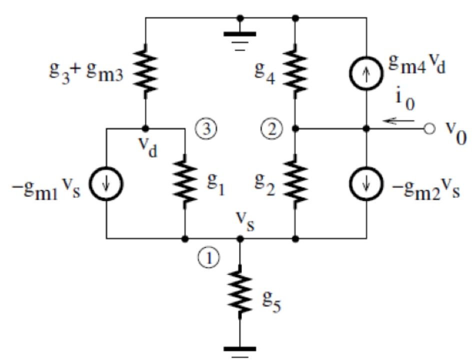
$$A_c = \frac{-g_5 g_{m1}}{g_1 g_5 + (g_3 + g_{m3})[g_5 + 2(g_1 + g_{m1})]}. \simeq \frac{-g_5}{2g_{m3}}$$

$$A_d = \left.\frac{v_0}{v_{id}}\right|_{v_{ic}=0} \qquad A_c = \left.\frac{v_0}{v_{ic}}\right|_{v_{id}=0}$$



larger the CMRR it is the better.

$$CMRR = \left|\frac{A_d}{A_c}\right| = \frac{\dfrac{g_{m1}}{g_1+g_3}}{\dfrac{g_5}{2g_{m3}}} \simeq 2\frac{g_{m1}g_{m3}}{g_5(g_1+g_3)}$$



$$r_0 = \frac{v_0}{i_0} = \frac{g_1g_5 + g_1(g_1+g_{m1}) + (g_3+g_{m3})[g_5 + 2(g_1+g_{m1})]}{(g_1+g_3)\{g_1g_5 + (g_3+g_{m3})[g_5+2(g_1+g_{m1})]\}}$$

where $g_1 = g_{ds_1}$, $g_3 = g_{ds_3}$, and $g_5 = g_{ds_5}$

$g_{m1}, g_{m3} \gg g_1, g_3, g_5,$ $\Rightarrow$ $\boxed{r_0 \simeq \dfrac{1}{g_1+g_3},}$

AN INTUITIVE METHOD OF SMALL SIGNAL ANALYSIS

Simplification of Small Signal Analysis

Small signal analysis is used so often in analog circuit design that it becomes desirable to find faster ways of performing this important analysis.

Intuitive Analysis (or Schematic Analysis)

Technique:

1.) Identify the transistor(s) that convert the input voltage to current (these transistors are called *transconductance transistors*).

2.) Trace the currents to where they flow into an equivalent resistance to ground.

3.) Multiply this resistance by the current to get the voltage at this node to ground.

4.) Repeat this process until the output is reached.

Simple Example:

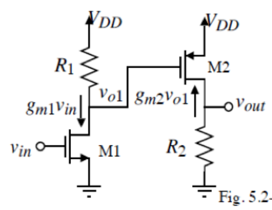
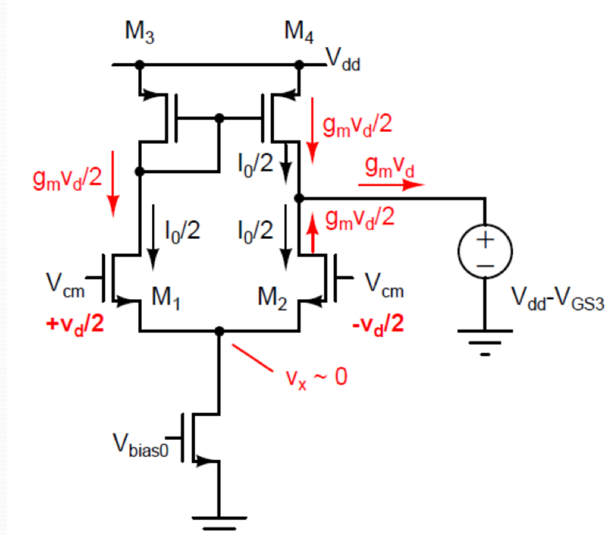


Fig. 5.2-10C

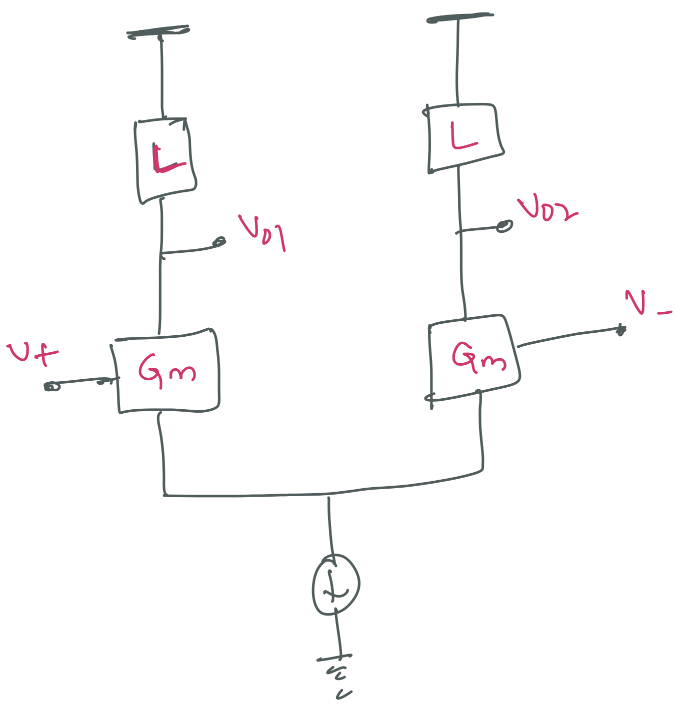
$v_{o1} = -(g_{m1}v_{in})\, R_1 \quad \rightarrow \quad v_{out} = -(g_{m2}v_{o1})R_2 \quad \rightarrow \quad v_{out} = (g_{m1}R_1 g_{m2}R_2)v_{in}$

Intuitive Analysis: Differential stage



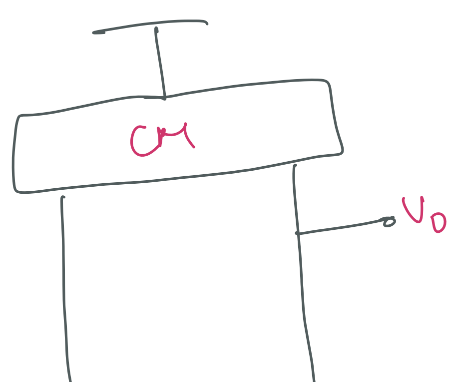
* our Idea is to find "Highest abstraction level" that is possible to analyze the differential pair -

→ So if we remember when we discussed "Diff. pair with Diff. o/p" we have sketched this scheme.

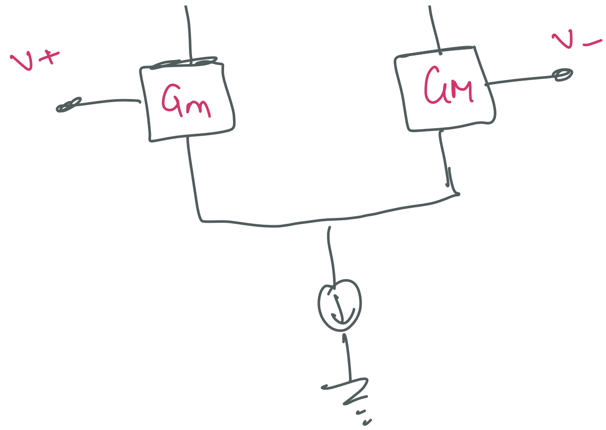


General Diff. pair with two o/p.

→ Differential pair with Current Mirror Load



Current Mirror allows us to have a single output voltage.



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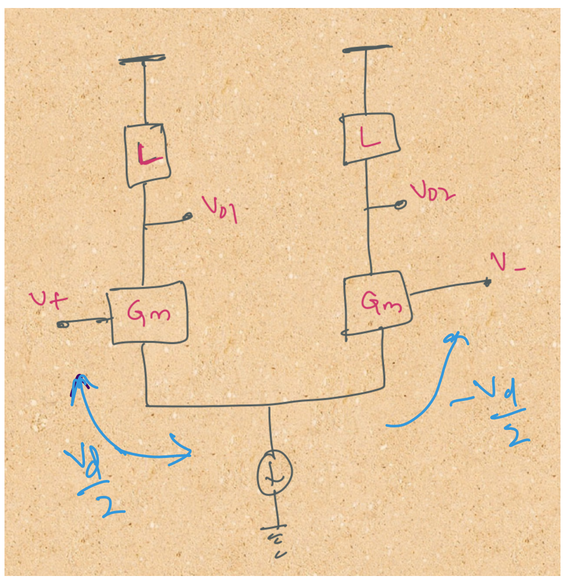
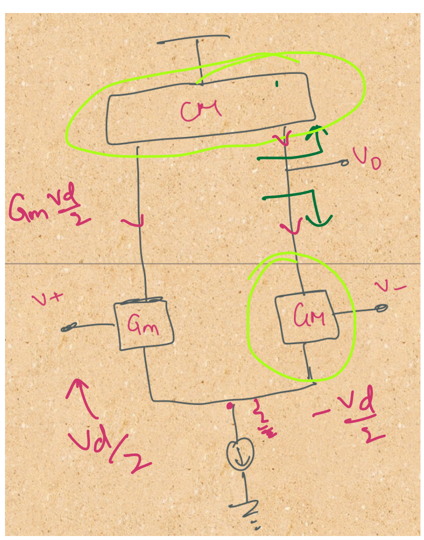


In this case we have V_{o1} which depends on $\frac{V_d}{2}$

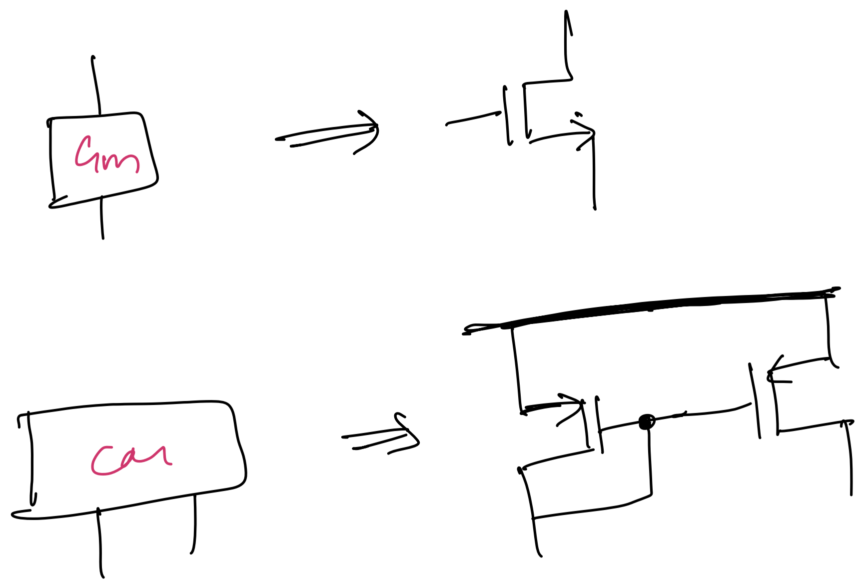


In this case

$$\boxed{\frac{V_o}{V_{id}} = G_m \cdot R_o}$$

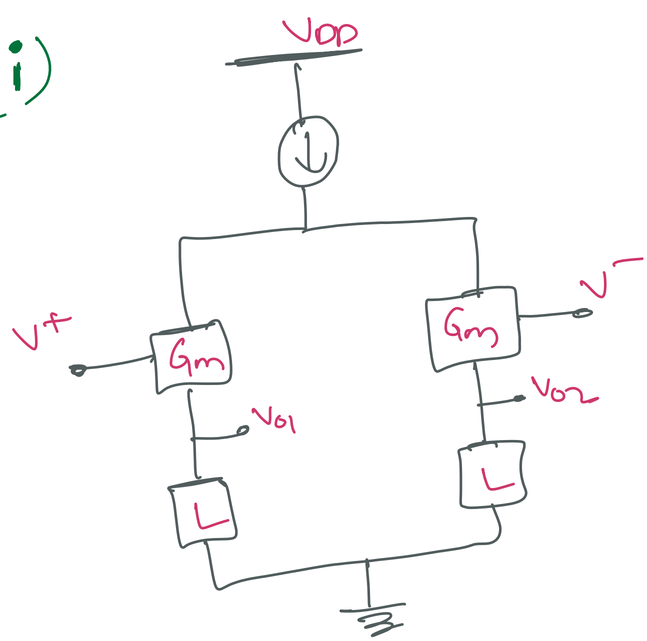
R_o = parallel of the o/p Resistance of CM & the Transconductance Branch

If substitute Gm Block with ~~Simple~~ Transistor, and Current Mirror Block with simple PMOS current mirror.

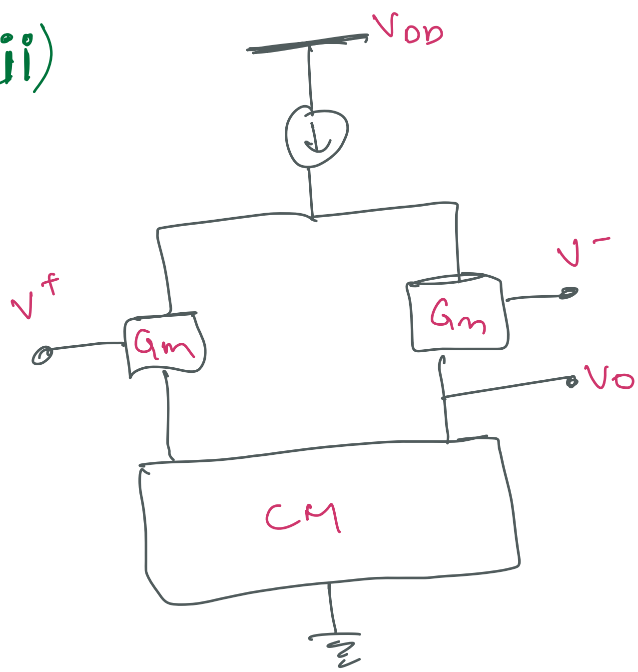


→ We can also have PMOS version of the Diff. pair.

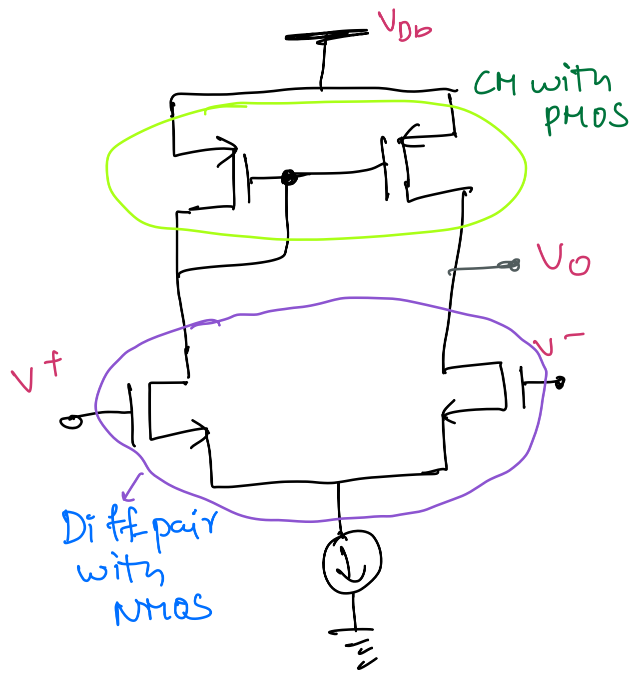
(i)

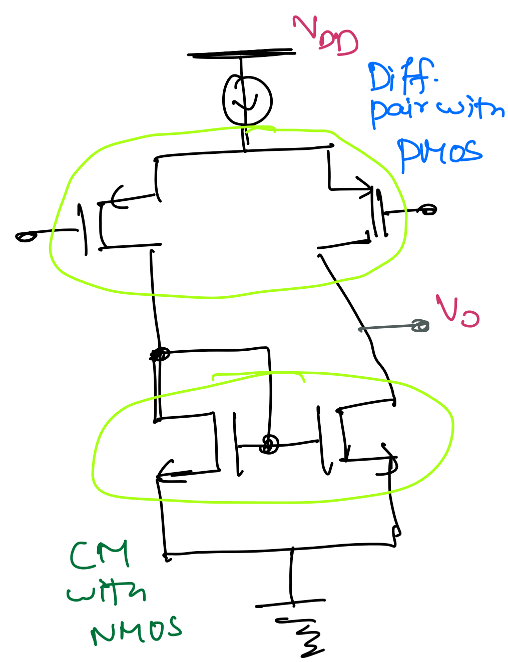


(ii)



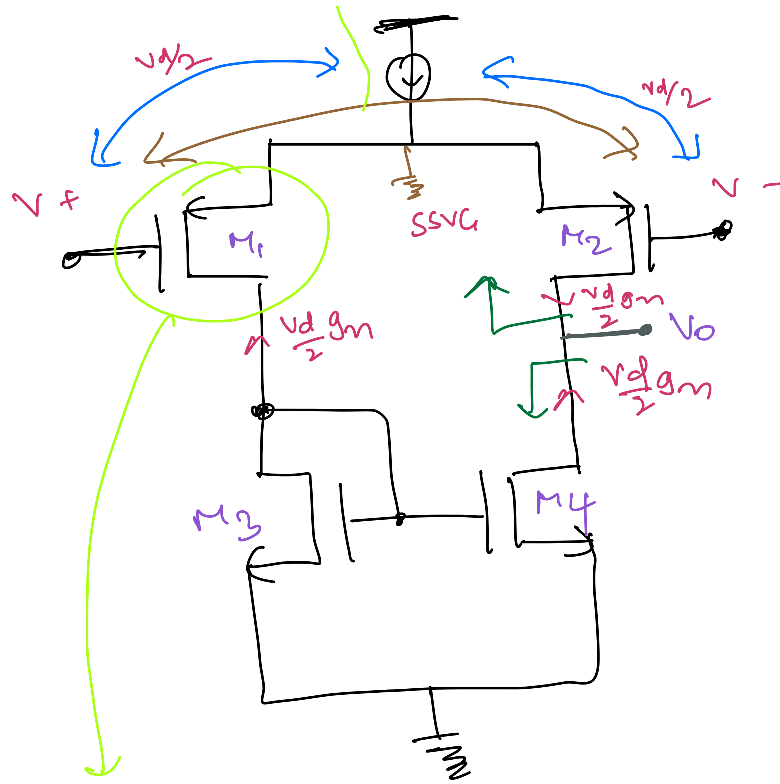
Implementation of Differential Pair with current mirror loading





Vd

V_{DD}



SSVG: Small signal virtual Ground

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→ This Transistor is converting the input voltage into current & the current is Fed as input of the current mirror.

More specifically if we take Differential input as the quantity we are interested in

→ $\frac{V_d}{2}$ is The Half of the differential voltage applied an is given as V^+

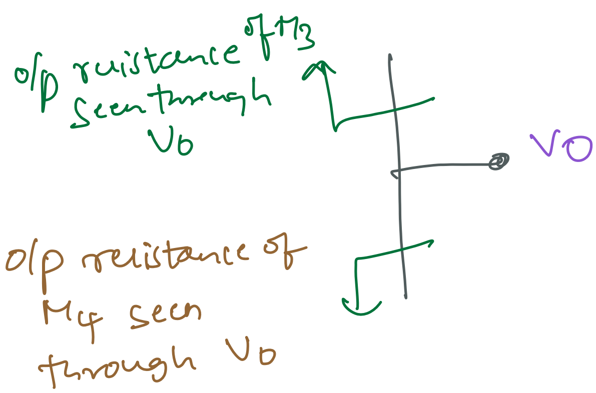
∴ The current through M Transistor is $i = \frac{V_d}{2} g_m$ which will act as an input current to the current Mirror which will try an mirror the current in the other branch.

& in the o/p Node we have current

$$I = \frac{V_d}{2} g_m \cdot 2$$

$$I = V_d g_m$$

$$V_o = I R_{out}$$
$$= I\left(\frac{1}{G_{out}}\right)$$
$$= I\left(\frac{1}{g_{m3}+g_{m4}}\right)$$



$$R_{out} = \frac{1}{g_{m3}+g_{m4}}$$

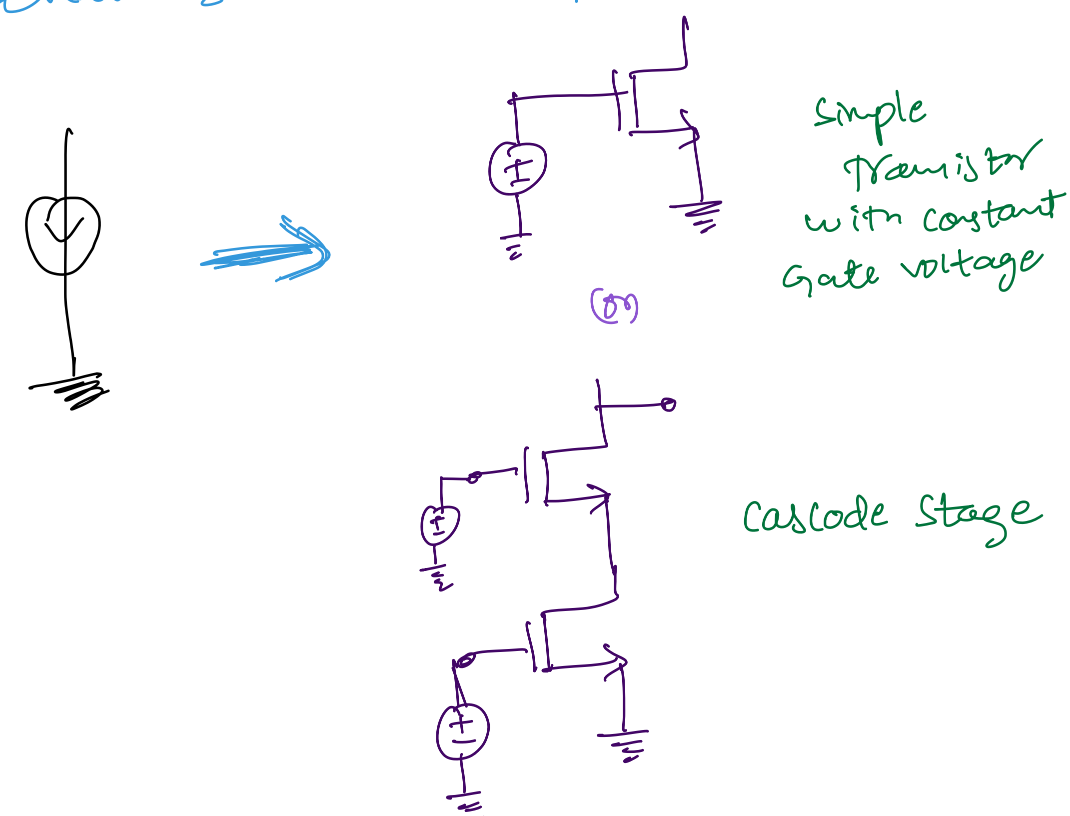
$$\therefore \text{Differential Gain} = \frac{V_o}{V_{id}} = \frac{I \Big/ \frac{1}{(g_{m3}+g_{m4})}}{V_{id}}$$

$V_{id} = V_d$

$$= \frac{V_d\, g_m / g_{m3}+g_{m4}}{V_d}$$

$$\boxed{\frac{V_o}{V_{id}} \approx \frac{g_m}{g_{m3}+g_{m4}}}$$

→ The current generator in Diff pair can be transformed into one of the scheme we have seen in the previous lessons.



∴ as we can see we have multiple options to use as current generators.

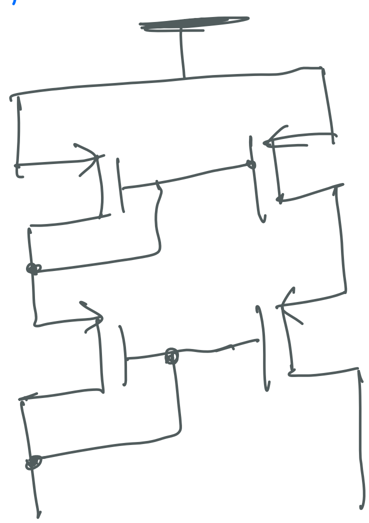
$$\frac{V_o}{V_{id}} = g_m \cdot R_o$$

$$R_o = \frac{1}{g_{m3} + g_{m4}}$$

→ Here we can see that the Differential gain is Directly proportional to the R_o (output Resistance) & g_m (Gain of the Transistor)

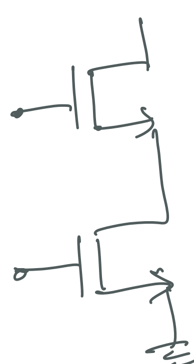
→ If we are capable of using large o/p Resistance & High g_m stage then we can achieve larger Differential Gain

$\boxed{CM}$ ⇒

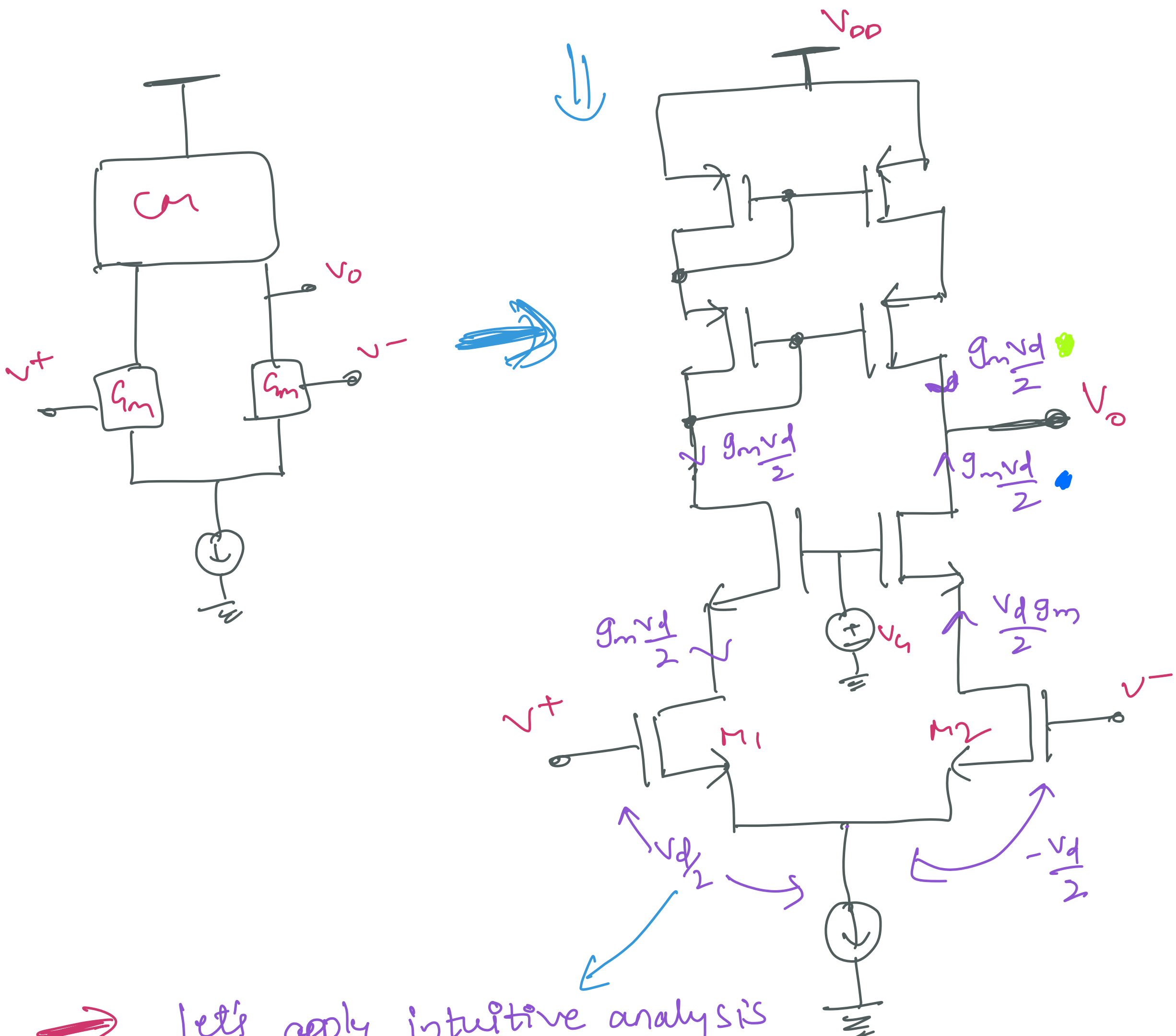


Cascode current mirror

$\boxed{G_m}$ ⇒



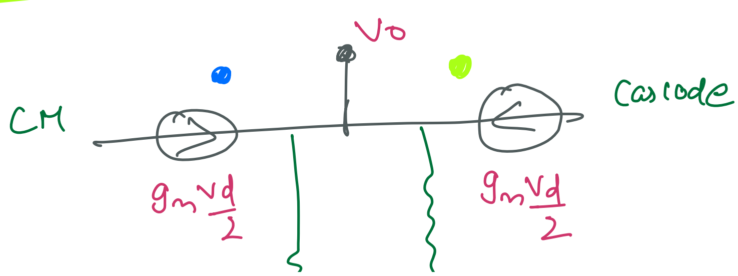
Cascode stage in Differential pair



let's apply intuitive analysis to the Topology

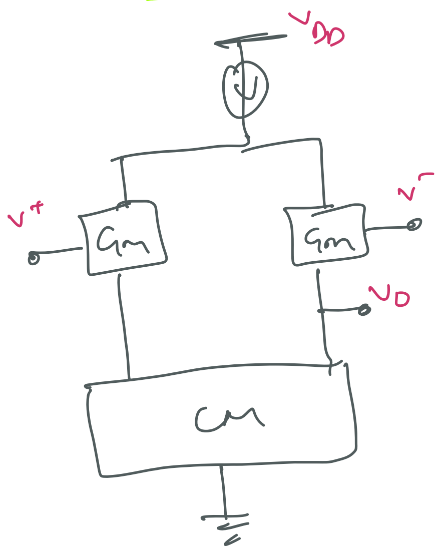
- Firstly The Trans conductance Transistors are M_1 & M_2

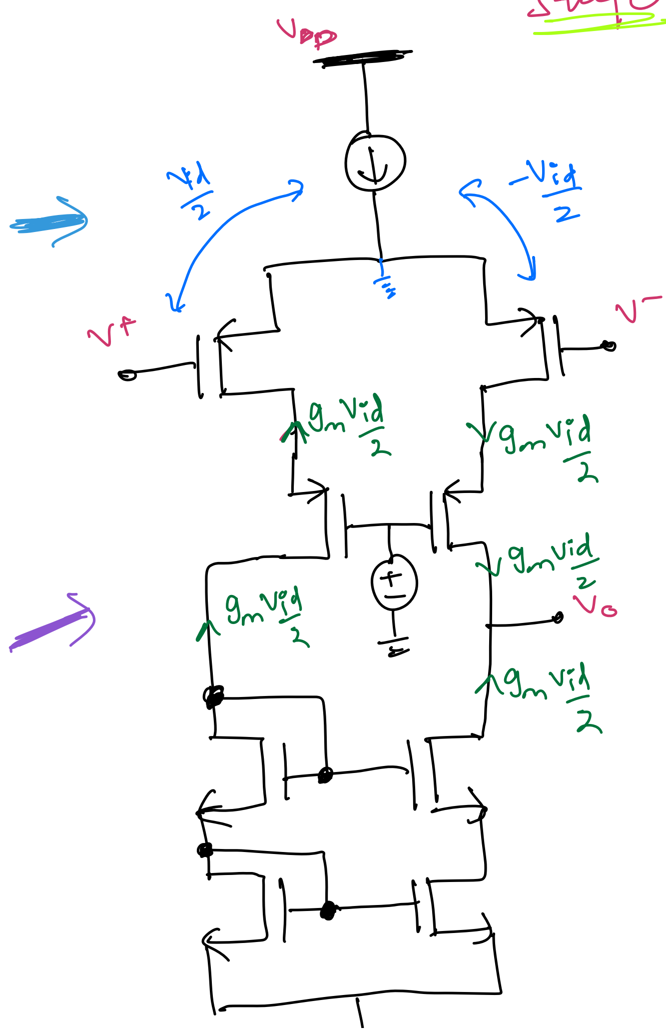
again at the o/p Node



$$\therefore \frac{V_o}{V_{id}} = \text{Differential Gain} = g_m \cdot R_{out} = g_m \cdot R_{out\,(casc \,||\, casc_{CM})}$$

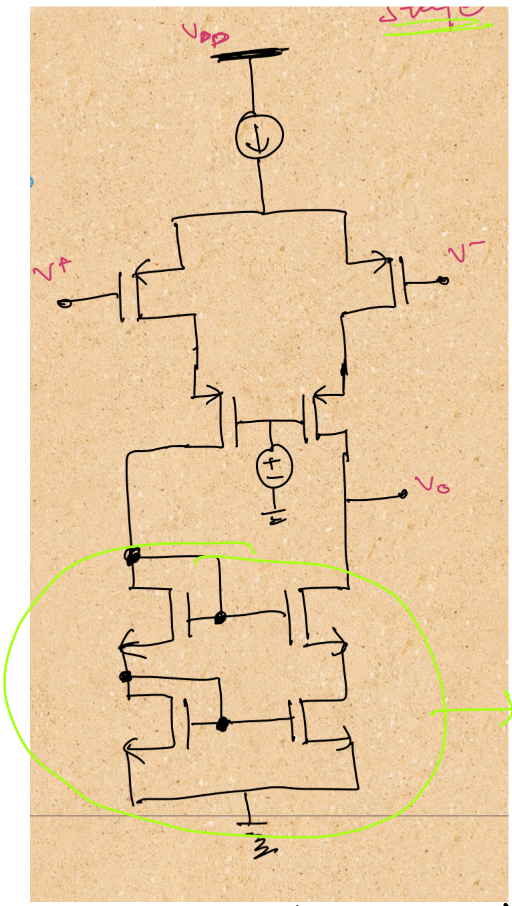
PMOS implementation of Diff pair using Cascode stage





Telescopic configuration

→ DC analysis of the above Topology to analyze the saturation points.



$V_{0min} = 2V_0 + V_T$

which is High

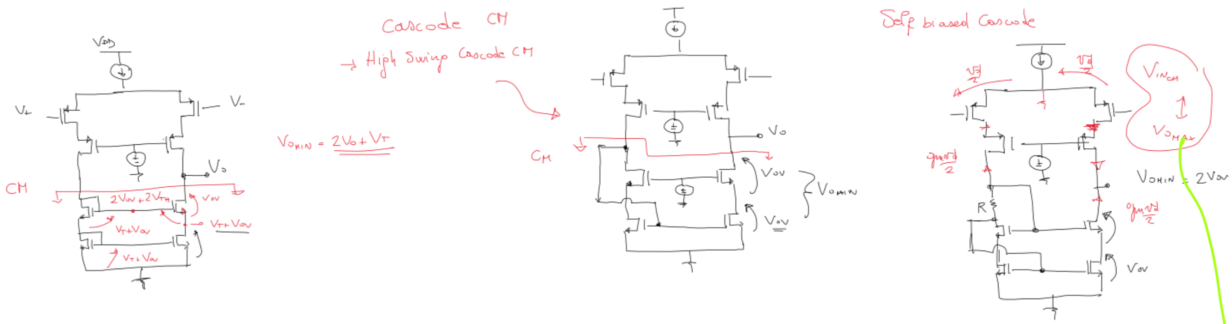
This current mirror can be implemented with another configuration

→ Q what need to be done to reduce V_{0min}?

use (i) Folded cascode CM

2) Highswing cascode CM

3) Self Biased cascode

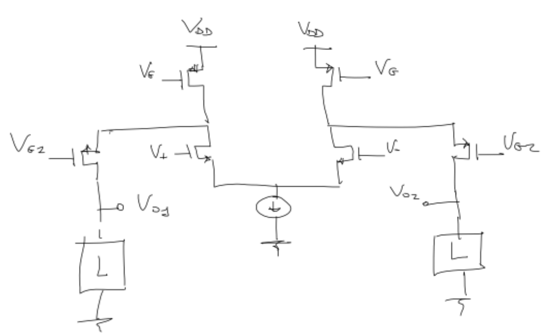


all the configurations we have studied earlier.

"V_{INCM} is always in conflict with V_{OMAX}"

***

Folded Cascode
Diff. Output



Folded Cascode
Single-ended Output
?

?

Q Folded Cascode with Single Ended output ?

by

Revanth Reddy Pannala

EBIT, Unibo